137

Tarea 18

Masa molar

1) Datos

$^{35}_{17}Cl = 75.53\%, 34.96 uma$ | $^{37}_{17}Cl = 24.47\% = 36.95 uma$

$\frac{75.53}{100} = 0.7553$ $\frac{24.47}{100} = 0.2447$

$34.96(0.7553) + 36.95(0.2447) = 35.44 \rightarrow$ masa atómica promedio

2) Datos

$^{63}Cu = 69.17\%, 62.92 uma$

$^{65}Cu = 30.83\%, 64.9278$

$0.6917(62.92) + 0.3083(64.9278) = 63.5456$
MAP masa atómica

Número de avogadro $= 6.022 \times 10^{23}$

1) $6.46 g He \left| \frac{1 mol He}{4.003 g He} \right. = 1.61 \rightarrow$ moles

2) $CO = 6 \times 10^{9} * \frac{1 mol}{6.022 \times 10^{23} CO} = 9.96 \times 10^{-15}$

3) $15.3 mol Au * \frac{196.96 g}{1 mol Au} = 3,013.49 g$

Masa molecular

H_2O, masa molecular

$H = 1.00794 * 2 = 2.01588$
$O = 15.9994 =$
18.01528 → masa molecular

CH_3OH

$C = 12.011$
$H = 4.03176$
$O = 15.9994$
32.04 → masa molecular

1) Datos ¿Cuantos moles hay en 6.67g de CH_4?
6.67g CH_4
$C = 12.011$
$H = 4.03176$
16.04276g

$6.7g \left| \frac{1 mol}{16.04 g} \right. = 0.42$

2) ¿Cuantos moles de átomos de calcio (Ca) hay en 77.4g de Ca?

$$77.4\,g\,Ca \left| \frac{1\,mol}{40.08\,g} \right. = 1.93\,mol$$

3) $[(CH_3)_2SO] = 78.13$

$C_2 = 24.022$
$H_6 = 6.05$
$S = 32.07$
$O = 15.99$

$$7.14\times10^3\,g \left| \frac{1\,mol}{78.13\,g} \right| \frac{2\,mol\,C}{1\,mol\,[(CH_3)_2SO]} \left| \frac{6.022\times10^{23}\,atom\,C}{1\,mol\,C} \right. = 1.10\times10^{26}\,atom\,C$$

$$7.14\times10^3\,g \left| \frac{1\,mol}{78.13\,g} \right| \frac{6\,mol\,H}{1\,mol\,[(CH_3)_2SO]} \left| \frac{6.022\times10^{23}\,atom\,H}{1\,mol\,C} \right. = 3.30\times10^{26}\,atom\,H$$

$$7.14\times10^3\,g \left| \frac{1\,mol}{78.13\,g} \right| \frac{1\,mol\,S}{1\,mol\,[(CH_3)_2SO]} \left| \frac{6.022\times10^{23}\,atom\,S}{1\,mol\,S} \right. = 5.50\times10^{25}\,atom\,S$$

Composición porcentual de los compuestos

$CHCl_3 = 119.38$

$C = 12.011 \div 119.38 = 0.10 * 100 = 10\ \%$

$H = 1.00794 \div 119.38 = \quad 0.84\%$

$Cl_3 = 106.36 \div 119.38 = 0.89 * 100 = \underline{89\%}$

100

Determinación experimental de fórmulas empíricas

H_2O ↑ F. empírica

H_4O_2 ↑ F. molecular

1) $C = 35.51\% - 35.51g * \frac{1mol}{12.011g} = 3mol$ 5

$H = 4.77\% - 4.77g * \frac{1mol}{1.00794} = 4.73mol$ 8

$O = 37.85\% - 37.85g * \frac{1mol}{15.9994} = 2.37mol$ 4

$N = 8.29\% - 8.29g * \frac{1mol}{14.0067} = 0.59$ 1

$Na = 13.60\% - 13.60g * \frac{1mol}{22.98977} = 0.59$ 1

$FE = C_5H_8O_4N_1Na_1$

$m = 169.11 \qquad 169 \div 169.11 = 1$

$FM = C_5H_8O_4NNa$

Gases

1) $Vol_1 = 3.20L$ $Vol_2 = 7.54L$
$Temp_1 = 125°C = 398$ $Temp_2 = ?$

$T_2 = \frac{V_1 T_1}{V_2}$ $T_2 = \frac{3.20\cancel{L} \cdot 398K}{7.54\cancel{L}} = 827°K$

2) Volumen en litros ocupado por 49.8g de HCl a TPE

$V = nRT/P$ $HCl = 49.8g = H = 1.00794$
$Cl = 35.4527$
36.44

$V = \frac{1.36 mol (0.0821L * atm/mol K) 273.15K}{1 atm}$

$48.8g \left| \frac{1 mol}{36.46g} \right. = 1.36 mol$

$V = 30.69L$